

Banking Industry: Structure and Competition

Chapter 11

Econ 351 – Monetary Economics

University of Tennessee, Knoxville

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Preview

This chapter examines the historical trends in the banking industry that help explain the unique structure of the U.S. system.

Three themes

- ▶ **History and regulation:** the dual banking system, branching restrictions, and the separation of banking from other financial services (Glass–Steagall) shaped a U.S. industry with unusually many banks.
- ▶ **Financial innovation:** banks and the shadow banking system respond to interest-rate volatility, information technology, and regulation (“loophole mining”), eroding traditional banking.
- ▶ **Consolidation and globalisation:** fewer, larger banks at home and the growth of international banking.

Outline

Historical Development of the Banking System

Financial Innovation & the Shadow Banking System

Structure of the U.S. Commercial Banking Industry

Bank Consolidation and Nationwide Banking

Separation of Banking and Other Financial Services

International Banking

Historical Development of the U.S. Banking System

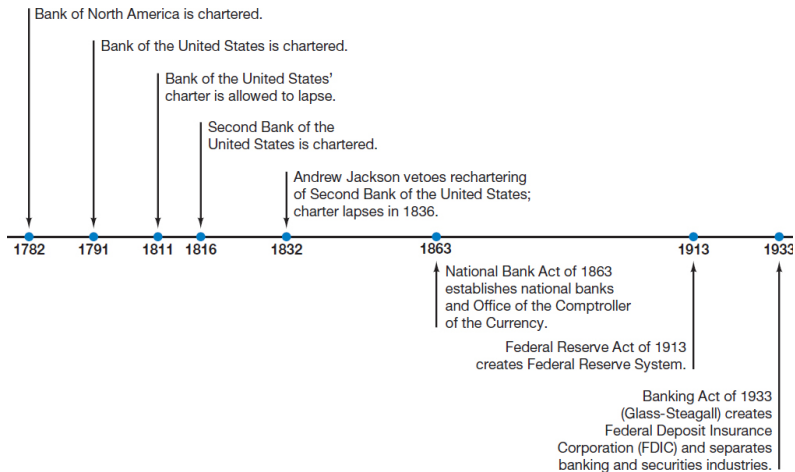
- ▶ **1782:** Bank of North America is chartered
- ▶ **Early debates:** Controversy over the chartering of banks
- ▶ **1863:** National Banking Act establishes federally chartered banks
 - ▶ Creates the Office of the Comptroller of the Currency (OCC)
 - ▶ Introduces the dual banking system
- ▶ **1913:** Creation of the Federal Reserve System



The First Bank of the United States, Philadelphia (chartered 1791), at the centre of the early debates over bank charters.

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Time Line of Early U.S. Commercial Banking



Time line of the early history of commercial banking in the United States.

Source: Mishkin, Figure 11.1

Primary Supervisors of U.S. Banks

- ▶ **Federal Reserve + State Banking Authorities**
 - ▶ Supervise state banks that are members of the Federal Reserve System
- ▶ **Federal Reserve**
 - ▶ Regulates bank holding companies
- ▶ **FDIC**
 - ▶ Supervises insured state banks that are not Fed members
- ▶ **State Banking Authorities**
 - ▶ Supervise state banks without FDIC insurance

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Financial Innovation

Motivation for Financial Innovation

- ▶ **Increase profits**
- ▶ Attract new customers and investors
- ▶ Bypass regulatory constraints

Financial Engineering

- ▶ The creation of new financial instruments and strategies is known as **financial engineering**.
- ▶ These innovations often respond to changes in the **economic and regulatory environment**.

Responses to Interest-Rate Volatility

Adjustable-rate mortgages

- ▶ Mortgage rates adjust with market interest rates
- ▶ Protect lenders when rates rise

Financial derivatives

- ▶ Financial contracts used to hedge interest-rate risk
- ▶ Payoffs depend on underlying assets

Information Technology and Financial Innovation

Improved information technology led to:

- ▶ New payment methods
 - ▶ Credit cards
 - ▶ Debit cards
- ▶ Electronic banking
 - ▶ ATMs
 - ▶ Online banking
- ▶ New financial markets
 - ▶ Junk bonds
 - ▶ Commercial paper

Securitization and the Shadow Banking System

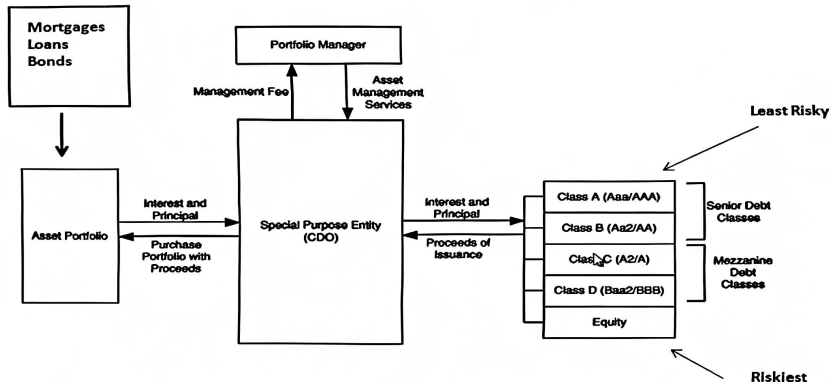
What is Securitization?

- ▶ Converting **illiquid financial assets** into **tradable securities**.
- ▶ Mortgages are bundled together and transformed into **mortgage-backed securities (MBS)**.

Role in the Shadow Banking System

- ▶ Loans are **pooled, packaged, and sold** to investors.
- ▶ Securitization expanded credit outside traditional banking.
- ▶ Played a major role in the **subprime mortgage crisis** in the mid-2000s.

Collateralized Debt Obligation (CDO)



How a CDO is built: mortgage loans are pooled by a special purpose vehicle and sliced into tranches sold to investors.

Source: web illustration (uncredited)

Avoidance of Existing Regulations

Loophole mining

- ▶ Reserve requirements act as a tax on deposits
- ▶ Limits on deposit interest encouraged disintermediation

Examples of innovations

- ▶ Money market mutual funds
- ▶ Sweep accounts

Regulation that raises the cost of doing business (reserve requirements, interest-rate ceilings) creates an incentive to innovate around it; “loophole mining” is financial innovation driven by the desire to bypass regulatory constraints.

Money Market Mutual Fund Panic of 2008

Background

- ▶ Pioneered by **Bruce Bent** in the 1970s.
- ▶ Provide investors with a safe place to hold short-term funds.

The 2008 Crisis

- ▶ Investors rapidly withdrew funds.
- ▶ One major fund “**broke the buck**”, triggering panic across the industry.

Government Response

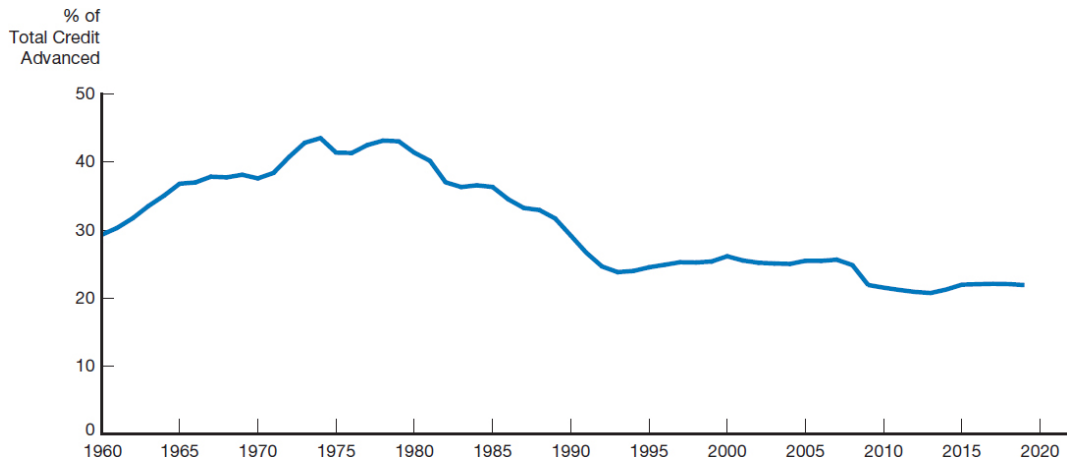
- ▶ Intervened to guarantee money market funds.
- ▶ The crisis led to calls for **stronger regulation** of these funds.

Financial Innovation and the Decline of Traditional Banking

- ▶ Banks' role as sources of funds has declined
- ▶ Banks' share of financial intermediation assets has fallen
- ▶ However, bank profitability has not necessarily declined
- ▶ Banks earn more income from **off-balance-sheet activities**

Traditional banking (taking deposits and making loans) has lost both its **cost advantage in acquiring funds** and its **income advantage in making loans**; the next two slides explain each.

Bank Share of Total Nonfinancial Borrowing



Bank share of total nonfinancial borrowing, 1960–2020.

Source: Mishkin, Figure 11.2 (*Federal Reserve Flow of Funds Accounts*)

Financial Innovation and the Decline of Traditional Banking

Decline in Cost Advantages for Funding

- ▶ Rising inflation increased interest rates and disintermediation
- ▶ Checkable deposits became less important as a low-cost funding source

Decline in Income Advantages for Lending

- ▶ Information technology reduced banks' role in short-term lending
- ▶ Lower transaction costs increased competition from other institutions

Financial Innovation and the Decline of Traditional Banking

Banks' Responses

- ▶ Expanded into riskier lending
 - ▶ Commercial real estate
 - ▶ Corporate takeovers and leveraged buyouts
- ▶ Increased **off-balance-sheet activities**
 - ▶ Growth of noninterest income

International Pattern

- ▶ Similar decline in traditional banking across other industrialized countries

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Restrictions on Branching

Restrictions on Branching

- ▶ **McFadden Act (1927)** limited interstate branching
- ▶ **State branching regulations** restricted banks within states

Responses by Banks

- ▶ Creation of **bank holding companies**
- ▶ Expansion of **automated teller machines (ATMs)**

Branching restrictions are one reason the United States has so many banks: a bank that could not open branches across state lines could not grow into a nationwide bank, so holding companies and ATMs became the loopholes.

Size Distribution of FDIC-Insured Banks

Assets	Number of Banks	Share of Banks (%)	Share of Assets Held (%)
Less than \$100 million	1,124	22.0	0.5
\$100 million–\$1 billion	3,168	61.9	6.7
\$1 billion–\$10 billion	680	13.3	10.0
\$10 billion–\$250 billion	131	2.6	32.3
More than \$250 billion	13	0.3	50.4
Total	5,116	100.00	100.00

Size distribution of FDIC-insured commercial banks by asset class.

Source: Mishkin, Table 11.1 (FDIC)

Ten Largest U.S. Banks, 2020

Bank	Assets (\$ trillions)	Share of All Commercial Bank Assets (%)
1. J.P. Morgan Chase & Co.	2.74	15.4
2. Bank of America Corp.	2.38	13.4
3. Citigroup Inc.	1.96	11.0
4. Wells Fargo & Co.	1.89	10.6
5. Goldman Sachs Group	0.93	5.2
6. Morgan Stanley	0.88	4.9
7. U.S. Bankcorp	0.48	2.7
8. PNC Financial Services Group	0.39	2.2
9. TD Group US	0.38	2.1
10. Capital One Financial Corp.	0.37	2.1
Total	12.40	69.6

Ten largest U.S. banks by assets, 2020.

Source: Mishkin, Table 11.2 (Federal Reserve)

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Bank Consolidation and Nationwide Banking

Trend

- ▶ The number of U.S. banks has declined sharply over the last 30 years

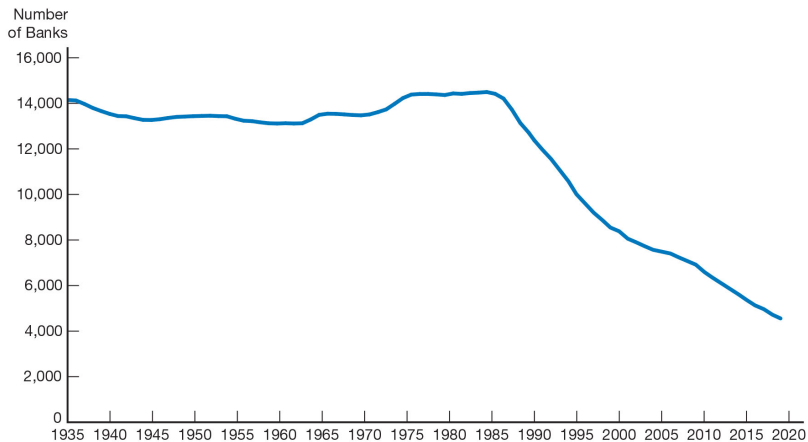
Causes of Consolidation

- ▶ Bank failures and mergers
- ▶ Deregulation: **Riegle-Neal Interstate Banking and Branching Efficiency Act (1994)**
- ▶ Economies of scale and scope from information technology

Result

- ▶ Fewer banks but much **larger institutions**

Number of Commercial Banks in the United States



Number of commercial banks in the United States, 1934–2020.

Source: Mishkin, Figure 11.3 (FDIC)

Future Structure of the U.S. Banking Industry

Current Situation

- ▶ The United States still has a large number of banks compared with other countries

Possible Future Trend

- ▶ The banking structure may gradually converge toward systems in Europe and Japan

Uncertainty

- ▶ The extent of this convergence remains debated

Bank Consolidation: Benefits and Costs

Benefits

- ▶ Competition removes inefficient banks
- ▶ Economies of scale and scope
- ▶ More diversified portfolios reduce failure risk

Costs

- ▶ Fewer community banks
- ▶ Less lending to small businesses
- ▶ Expansion into new areas may increase risk

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Separation of Banking and Regulation in US

Separated Financial Structure

- ▶ Different types of financial institutions operate separately
- ▶ Different regulators oversee different sectors

Advantages

- ▶ Specialization in regulation
- ▶ Clear division of responsibilities

Disadvantages

- ▶ Regulatory gaps
- ▶ Overlapping supervision

Glass-Steagall Act (1933)

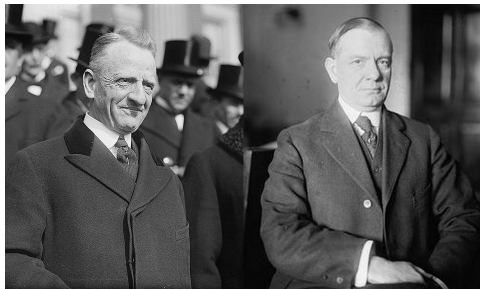
Separation of **commercial banking** and **investment banking**

Restrictions

- ▶ Commercial banks prohibited from underwriting corporate securities
- ▶ Banks restricted from brokerage and investment banking activities

Later Developments

- ▶ Section 20 loophole allowed some securities activities through affiliates
- ▶ U.S. Supreme Court validated this interpretation in 1988



Senator Carter Glass (left) and Representative Henry Steagall, sponsors of the 1933 Act.

Wikimedia Commons, Public domain

Gramm-Leach-Bliley Act (1999)

Main Change

- ▶ Effectively repealed key provisions of the **Glass-Steagall Act**
- ▶ Allowed integration of **banking, securities, and insurance**

Regulatory Structure

- ▶ **States** regulate insurance activities
- ▶ **SEC** oversees securities activities
- ▶ **Federal Reserve** supervises bank holding companies

Impact

- ▶ Encouraged consolidation in the banking industry

Banking and Financial Services Around the World

- ▶ Some countries do not separate banking and securities industries: Germany, Holland, Switzerland
- ▶ Universal banking in Great Britain
 - ▶ Banks can engage in securities underwriting
- ▶ Japanese-style banking system



Deutsche Bank's twin towers in Frankfurt: a German universal bank combining commercial and investment banking.

Wikimedia Commons, Paul Colin Hennig, CC BY-SA

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The Global Financial Crisis and Investment Banks

- ▶ After the weakening of the Glass-Steagall separation, financial services moved toward larger and more complex institutions
- ▶ During the 2008 financial crisis, the traditional investment banking model collapsed
- ▶ Between March and September 2008, all five major U.S. investment banks disappeared or were transformed

The 2008 crisis ended the era of the large stand-alone U.S. investment bank: the survivors became bank holding companies supervised by the Federal Reserve, or were absorbed by commercial banks.

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International Banking and the Eurodollar Market

Rapid growth of international banking, driven by

- ▶ Expansion of global trade
- ▶ Growth of multinational corporations
- ▶ Profitable global investment banking
- ▶ Access to Eurodollar markets

Eurodollar Market

- ▶ Dollar-denominated deposits held outside the United States
- ▶ Widely used currency in international trade
- ▶ Offshore deposits are not subject to U.S. regulations
- ▶ Important funding source for U.S. banks

U.S. Banks Overseas and Foreign Banks in the U.S.

Structure of U.S. Banking Overseas

- ▶ Shell branches
- ▶ Edge Act corporations
- ▶ International Banking Facilities (IBFs)
 - ▶ Not subject to certain U.S. regulations and taxes
 - ▶ Cannot lend to domestic residents

Foreign Banks in the United States

- ▶ Agency office
 - ▶ Can lend and transfer funds
 - ▶ Cannot accept deposits from U.S. residents
- ▶ Subsidiary U.S. bank
 - ▶ Subject to U.S. regulations
 - ▶ Owned by a foreign bank

Largest Banks in the World, 2020

Bank	Assets (U.S. \$ trillions)
1. Industrial and Commercial Bank of China, China	4.32
2. China Construction Bank Corp., China	3.82
3. Agricultural Bank of China, China	3.70
4. Bank of China, China	3.39
5. JPMorgan Chase, US	3.14
6. HSBC Holdings plc, United Kingdom	2.92
7. Mitsubishi UFJ Financial Group, Japan	2.89
8. Bank of America, US	2.62
9. BNP Paribas	2.43
10. Credit Agricole Group, France	1.98

Ten largest banks in the world by assets, 2020.

Source: Mishkin, Table 11.3

Chapter Summary

- ▶ The U.S. dual banking system (state and national charters, multiple regulators) dates from the National Banking Act of 1863 and the creation of the Fed in 1913.
- ▶ Financial innovation responds to interest-rate volatility, information technology, and regulation; securitization and the shadow banking system grew outside traditional banking.
- ▶ Traditional banking declined as banks lost cost advantages in funding and income advantages in lending, pushing them into riskier lending and off-balance-sheet activities.
- ▶ Branching restrictions produced an industry with thousands of banks; deregulation (Riegle–Neal 1994, Gramm–Leach–Bliley 1999) drove consolidation and the integration of banking, securities, and insurance.
- ▶ International banking grew with global trade, multinational corporations, and the Eurodollar market.